

# Zhijie Zheng

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Github | Google Scholar

## EDUCATION

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|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>University of California, Davis</b><br>Ph.D. in Computer Science (Advisor: Dr. Jiawei Zhang)   GPA: 3.9/4.0 | Davis, CA<br>Sept 2025 - June 2030 (Expected) |
| <b>Sun Yat-sen University</b><br>M.S. in Electronic Science   GPA: 3.9/4.0                                     | Guangzhou, China<br>June 2025                 |
| <b>Sun Yat-sen University</b><br>B.S. in Electronic Science   GPA: 3.9/4.0                                     | Guangzhou, China<br>June 2022                 |

## PUBLICATION

- TTSA3R: Training-Free Temporal-Spatial Adaptive Persistent State for Streaming 3D Reconstruction.**  
Under Review at ECCV 2026. [Paper link](#) [Project github](#)
- UniPT: A Unified Representation Pre-Training for Multi-dataset 3D Object Detection.**  
PRICAI 2025. [Paper link](#) [Project github](#)
- RAFDet: Range View Augmented Fusion Network for Point-Based 3D Object Detection.**  
IEEE Transactions on Multimedia. [Paper link](#) [Project github](#)
- DTSSD: Dual-Channel Transformer-Based Network for Point-Based 3D Object Detection.**  
IEEE Signal Processing Letters. [Paper link](#)

## EXPERIENCE & PROJECTS

- Research on Multi-view Streaming 3D Reconstruction Algorithm | UC Davis** Oct 2025 - Feb 2026
- Developed a training-free framework to maintain reconstruction accuracy in streaming multi-view 3D scene modeling.
  - Designed adaptive mechanisms through joint temporal-spatial analysis to mitigate performance degradation.
  - Tested on six datasets and maintained only  $1.33\times$  error increase versus  $4\times$  accuracy degradation in the baseline method.
- Research on Multi-dataset 3D Object Detection Algorithm | Beijing Novauto** March 2024 - Oct 2024
- Proposed a pre-training network to learn representations across datasets to alleviate the dataset discrepancy challenges.
  - Implemented self-supervised learning with masked autoencoding using PyTorch and validated on three benchmarks.
  - Achieved improvements of 2.29%, 1.45% and 2.08% on Waymo, nuScenes and KITTI over training-from-scratch.
- Research on Multi-view Enhanced 3D Object Detection Algorithm | SYSU** Aug 2023 - Dec 2023
- Built a multi-view fusion network to address the issue of insufficient representation of sparse point clouds.
  - Integrated point cloud representations with cylindrical range view and cross-view attention mechanisms.
  - Improved detection accuracy by 4.2% mAP on Waymo and 1.99% on KITTI datasets over baseline.
- Research on Dual-channel Feature Enhanced 3D Object Detection | SYSU** Aug 2022 - March 2023
- Developed the dual-channel Transformer architecture integrating PointNet++ and Transformer to address insufficient point cloud feature learning.
  - Conducted studies on industry-standard KITTI dataset and achieved 4.18% detection improvement over baseline.
  - Collaborated with research team and documented technical findings for publication.

## TECHNICAL SKILLS

**Programming Languages:** Python, C/C++, Shell, CUDA  
**Deep Learning & CV Frameworks:** PyTorch, OpenPCDet, OpenCV, Open3D, Point Cloud Library (PCL)  
**Tools & Platforms:** Linux (Ubuntu), Docker, Git, VS Code, PyCharm, VMWare, Robot Operating System  
**Language Skills:** Chinese Mandarin (native), English (fluent)

## HONORS & AWARDS

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|-------------------------------------------------------------------------|-----------|
| National Scholarship for Master's Students in 2023                      | Dec 2023  |
| Second Prize in 2023 China College IC Competition South China           | Aug 2023  |
| Excellent Undergraduate Degree Thesis of Sun Yat-sen University in 2022 | June 2022 |